Nilava Metya

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Introduction

I identify as an applied mathematician. My research encompasses applying techniques from algebra and geometry to algorithmic and statistical problems in *optimization* and *machine learning* from a structural perspective. More specifically, I like to borrow tools from algebra, convex geometry and probability for machine learning.

Education

Rutgers, the State University of New Jersey - New Brunswick — **Sep '22 – (expected) '27**
Doctor of Philosophy in Mathematics — *Piscataway, New Jersey, USA*
(passed written qualifying exams in first attempt just before program started)

Chennai Mathematical Institute — **Aug '19 – May '22**
Bachelor of Science (Honours) in Mathematics and Computer Science | ***CGPA:*** 9.72/10 — *Chennai, Tamil Nadu, India*
Position: ***Third*** *(out of* 55 *students)*
(completed degree requirements in 2.5 *years)*

Don Bosco School, Liluah — **Apr '06 – Mar '19**
Indian School Certificate (ISC) *2019* | ***Percentage:*** 97.25% — *Howrah, West Bengal, India*
Position: ***First*** *in science stream (∼* 55 *students),* ***second*** *overall (∼* 180 *students)*
Indian Certificate of Secondary Education (ICSE) *2017* | ***Percentage:*** 96.6%
Position: ***First*** *in school (∼* 180 *students)*

Current and Upcoming Research

9. *N Metya, S Mukherjee. Sample complexity of identifying linear dynamical systems. 2026+. *Upcoming*
8. N Metya, O Leong, Y S Soh. Convex optimization over the space of convex bodies. 2026+. *Upcoming*
7. *N Metya, J Saunderson. A new Hyperbolic Chernoff bound. 2026+. *Upcoming*
6. N Metya. ML degrees via chordal completion. 2026.
5. *N Metya, S Mukherjee. Approximate Uniformity in Finite Convolution Models. 2026.
4. *S Gaungly, N Metya, A Morozov, A Sengupta. Activation Functions Control Finite-Width Concentration in Wide Neural Networks. 2026.
   4th Workshop on High-dimensional Learning Dynamics at ICML 2026.
3. M Smart, S Gaungly, N Metya, A Morozov, A Sengupta. Attention as In-Context Empirical Bayes: A Two-Stage View via Particle Dynamics. 2026.
   NeurIPS 2026 and ICML 2026 Workshop on Foundations of Deep Generative Models.
2. N Metya, A Sinha. Temporal robustness in discrete time linear dynamical systems. 2025.
   Submitted
1. *G DePaul, S Hoşten, N Metya, I Nometa. Degrees of the Wasserstein distance to small toric models. 2024.
   Journal of Algebraic Statistics

*Alphabetical.

Talks

Monash University **Jan '27**
Maximum Likelihood Degrees via One-Edge Chordal Completion.

Young Mathematician Forum at the Department of Mathematics, NUS **Jan '27**
Maximum Likelihood Degrees via One-Edge Chordal Completion.

Chennai Mathematical Institute **Jan '27**
Maximum Likelihood Degrees via One-Edge Chordal Completion.

Department of Mathematics, Indian Institute of Science **Jan '27**
Maximum Likelihood Degrees via One-Edge Chordal Completion.

School of Mathematical Sciences, Shanghai Jiao Tong University **Dec '26**
Maximum Likelihood Degrees via One-Edge Chordal Completion.

Rutgers Algebra Seminar **Oct '26**
Maximum Likelihood Degrees via One-Edge Chordal Completion.

INFORMS invited session on AI for Security and Resilience in Complex Networks **Oct '25**
Temporal robustness in discrete time linear dynamical systems.

ACMU, Indian Statistical Institute **Aug '25**
Temporal robustness in discrete time linear dynamical systems.

School of Computing, National University of Singapore **May '25**
Temporal robustness in discrete time linear dynamical systems.

Mina Rees Women and Math Conference, CUNY **Mar '25**
Degrees of the Wasserstein distance to small toric models.

Academic Visits

Monash University | *Host: James Saunderson* **Jan '27**
Department of Mathematics, National University of Singapore | *Host: Yong Sheng Soh* **Jan '27**
Shanghai Jiao Tong University | *Host: Oscar Leong* **Dec '26**
School of Computing, National University of Singapore | *Host: Jonathan Scarlett* **May-Aug '26**
Department of Mathematics, National University of Singapore | *Host: Yong Sheng Soh* **May-Aug '26**
School of Computing, National University of Singapore | *Host: Jonathan Scarlett* **May-Aug '25**
Department of Mathematics, Sorbonne University | *Host: Anne Marie Aubert* **May '22**

Attendance in conferences/workshops

Annual Summer School on Mathematical Aspects of Data Science | *Summer School* | NUS, Singapore **Jul '26**
Princeton Machine Learning Theory Summer School | *Summer School* | Princeton **Aug '24**
Efficient Algorithms for High Dimensional Metrics | *Workshop* | DIMACS, Rutgers, New Brunswick **May '24**
Bayesian Statistics and Statistical Learning | *Workshop* | IMSI, Chicago **Dec '23**
Algebraic Statistics for Ecological and Biological Systems | *Workshop* | IMSI, Chicago **Oct '23**
Apprenticeship Week: Varieties from Statistics | IMSI, Chicago **Oct '23**
Invitation to Algebraic Statistics and Applications | IMSI, Chicago **Sep '23**
Algebraic Methods in Biochemical Reaction Networks | MPI, Leipzig **Jun '23**
Joint Mathematics Meetings | Boston **Jan '23**
AlGeCom-XII (Algebra Geometry and Combinatorics day) | UIUC **Oct '22**

Teaching and Grading

Workshop leader for Calculus II | Rutgers **Sep – Dec '23, Jan – Apr '24**

Instructor – PhD written qualifying exam in Algebra | Rutgers University **Jun – Aug '23**

Head Counselor at PROMYS India | IISc Bangalore **May – Jun '23**

Counselor at PROMYS | Boston University **Jul – Aug '20, '21**

Grader | Rutgers University

Combinatorial methods in Complexity Theory	Sep – Dec '26
Graduate Algebra I	Sep – Dec '25
Linear Algebra and Applications	Jan – Apr '25
Mathematical Statistics	Sep – Dec '24
Algebra II	Jan – Apr '24
Linear Algebra and Applications	Sep – Dec '23
Analysis II	Jan – Apr '23
Topics in Applied Algebra	Jan – Apr '23
Topology	Sep – Dec '22
Theory of Numbers	Sep – Dec '22

Teaching Assistant | Chennai Mathematical Institute

Algebra II (Group theory)	*BSc 1st year*	***Prof Manoj Kummini***	Jan – May '22
Algebra I (Linear algebra)	*BSc 1st year - head tutor*	***Prof T R Ramadas***	Sep – Dec '21
Functional Programming in Haskell	*BSc and MSc Comp. Sci. 1st year*	***Prof S P Suresh***	Sep – Dec '21
Probability Theory	*BSc 1st year*	***Prof P Sankaran***	Apr – Jul '21
Discrete Mathematics	*BSc 1st year*	***Prof K V Subrahmanyam***	Apr – Jul '21
Design and Analysis of Algorithms	*MSc Data Science 1st year*	***Prof G Philip***	Apr – Jul '21
Algebra I (Linear algebra)	*BSc 1st year*	***Prof T R Ramadas***	Dec '20 – Mar '21
Functional Programming in Haskell	*BSc and MSc Comp. Sci. 1st year*	***Prof S P Suresh***	Dec '20 – Mar '21

Honours and Awards

Nominated by Rutgers Math department for SLMath summer school **Jun '23**
Summer school at Leipzig - awarded full travel funding. Only two students from Rutgers Math were fully funded by SLMath.

Academic Excellence Award at Rutgers **Sep '22**
Received a certificate and $100 based on performance in Written Qualifying Exams.

Shriram Scholarship at CMI **'19 – '22**
Received institutional fee waiver and monthly stipend (based on entrance exam).

Ranked 4th nationally at the *Bachelor of Statistics* (B.Stat.) entrance examination **'19**
Indian Statistical Institute (ISI)

Informatics Olympiad **'17, '18, '19**
*Selected among (approx) top 100-130 school students in India in **Zonal Informatics Olympiad** (ZIO).*

Mathematical Olympiad **Jan '18**
*Selected for **Indian National Mathematical Olympiad** (INMO) **Training Camp** | top 30 school students in West Bengal.*

Program in Mathematics for Young Scientists (PROMYS) **'18, '19, '20, '21**
*Awarded the **Tara and Jasubhai Mehta Fellowship** to PROMYS (among 5 Indian school students in 2018) based on a competitive process. Participated twice as a student ('18, '19) and twice as a counselor ('20, '21).*

Others

- Qualified for **International Collegiate Programming Contest** (ICPC) Kharagpur regionals and Amritapuri regionals in 2019 and secured rank 35 among (approx) 90 university teams at Kharagpur.
- Selected among top 30 students in India to participate in **Scholastic Test of Excellence in Mathematical Sciences** (STEMS) camp at CMI in 2018, based on a competitive exam (across grades $9 - 12$ and across Math, Physics, Computer Science).
- Secured the **third position** in **Mathematics Talent Reward Programme** (MTRP) 2016, organized by ISI Kolkata, based on a competitive exam and quizzes at a camp.

Service

Coffee Hour (departmental socializing) | *Coordinator and Volunteer* | Rutgers Math Department **Jan '25 - Apr'27**

Physics of Learning Theory (PhyLT) | *Organizer* | Rutgers Math Department **Jan '25 - Apr'26**

Algebra 'N' Geometry Learning Seminar (ANGeLS) | *Organizer* | Rutgers Math Department **Jan - Apr '23**

Student Seminar | *Organizer* | Chennai Mathematical Institute **Oct - Dec '22**

ICO Camp (online) | *Combinatorics teacher* | CodeChef **Nov '20**

Selected Coursework

- Quantum Computation
- Matrix Computations
- Statistics with R
- Algebraic Topology
- Complex Analysis
- Quiver Representations
- Algebraic Number Theory
- Haskell
- Algorithm Design + Analysis
- Algebraic Geometry^{Princeton}
- Homological Algebra^{Rutgers}
- Probability^{Rutgers}
- Data Mining^{Rutgers}
- Topological Data Analysis^{Rutgers}
- Convex Optimization^{Princeton}
- Convexities in Banach Spaces^{Princeton}
- Random Matrices^{Princeton}

Skills

Languages	Bengali (mother tongue), English (fluent), Hindi (fluent)
Programming	JAVA, C++, Python, Haskell, R, HTML, SageMath, Macaulay2, MATLAB
Documentation	LaTeX, Microsoft Word